

1. AWS Free Tier Limits & Zero-Cost Strategy

This deployment is specifically optimized to stay **100% within the AWS Free Tier** limits without incurring unexpected credit card charges:

AWS Service	Free Tier Allowance	Our Configuration	Cost
EC2 Compute	750 hours/month of t2.micro or t3.micro (1 vCPU, 1GB RAM)	Single t2.micro (or t3.micro) running 24/7 with 4GB Virtual Swap memory	\$0.00 / FREE
EBS Storage	30 GB/month of General Purpose SSD (gp3/gp2)	Exactly 30 GB gp3 SSD volume	\$0.00 / FREE
S3 Storage	5 GB Standard storage, 20,000 GETs, 2,000 PUTs/month	WebP image optimization (~200KB/image = ~25,000 images capacity)	\$0.00 / FREE
CloudFront CDN	1 TB/month Data Transfer Out + 10,000,000 requests (Always Free!)	Caches all images at global edge servers	\$0.00 / FREE
ACM SSL Cert	Unlimited public SSL certificates for custom domains	Free wildcard certificate for *.gowildkarunadu.online	\$0.00 / FREE
DNS Routing	Hostinger DNS Zone Editor included with your domain	Manage DNS in Hostinger DNS Editor (saves \$0.50/mo Route 53 fee)	\$0.00 / FREE

2. Step 1: Configure Free DNS in Hostinger

Cost-Saving Tip: Use Hostinger DNS (100% Free)

AWS Route 53 charges \$0.50/month per hosted zone. By managing your DNS records directly inside Hostinger's free DNS Zone Manager, your monthly DNS cost is **\$0.00**.

1. Log into your **Hostinger Dashboard (hPanel)**.
2. Go to **Domains** → click **gowildkarunadu.online** → click **DNS / Nameservers**.
3. Ensure nameservers are set to **Hostinger default nameservers**.
4. Scroll down to the **Manage DNS records** section and add these records (replace YOUR_EC2_IP with your

server's public IP from Step 3):

Type	Name	Points To / Content	TTL
A	@	YOUR_EC2_PUBLIC_IP	300
CNAME	www	gowildkarunadu.online	300
A	admin	YOUR_EC2_PUBLIC_IP	300
A	api	YOUR_EC2_PUBLIC_IP	300
A	admin-api	YOUR_EC2_PUBLIC_IP	300
CNAME	images	your-id.cloudfront.net (from Step 4)	300

3. Step 2: Launch Free Tier EC2 Server with 4GB Swap

3.1 Launch Instance

1. Open **AWS Console** → **EC2** → **Launch Instance**.
2. **Name:** gowild-karunadu-prod
3. **OS:** **Ubuntu Server 24.04 LTS (HVM)**, 64-bit (x86).
4. **Instance Type:** Select t2.micro (or t3.micro in Mumbai) labeled "**Free tier eligible**".
5. **Key Pair:** Create and download gowild-key.pem.
6. **Storage:** Set to **30 GiB gp3** (100% Free tier eligible).
7. **Security Group Inbound Rules:**
 - SSH (Port 22) → 0.0.0.0/0
 - HTTP (Port 80) → 0.0.0.0/0
 - HTTPS (Port 443) → 0.0.0.0/0
8. Click **Launch Instance**. Copy the **Public IPv4 Address**.

3.2 Connect via SSH & Configure 4GB Swap Memory

Crucial for Free Tier (1GB RAM)

A t2.micro instance has 1GB physical RAM. Running MySQL 8 and two Node.js APIs can cause out-of-memory errors unless you enable a **Swap File**. This creates 4GB of virtual memory on your SSD, ensuring 100% stability at zero extra cost.

```
# SSH into the server

chmod 400 gowild-key.pem

ssh -i gowild-key.pem ubuntu@YOUR_EC2_PUBLIC_IP


# Create a 4GB Swap file

sudo fallocate -l 4G /swapfile

sudo chmod 600 /swapfile

sudo mkswap /swapfile

sudo swapon /swapfile


# Make swap permanent across reboots

echo '/swapfile none swap sw 0 0' | sudo tee -a /etc/fstab


# Verify swap is active (should show ~4GB Swap)

free -h
```

4. Step 3: Install Free Server Software & MySQL

```
# Update OS packages

sudo apt update && sudo apt upgrade -y


# Install Node.js 20 LTS

curl -fsSL https://deb.nodesource.com/setup_20.x | sudo -E bash -

sudo apt install -y nodejs build-essential git nginx certbot python3-certbot-nginx mysql-server


# Install PM2 globally

sudo npm install -g pm2


# Configure MySQL
```

```
sudo mysql
```

4.2 Create Database & Import Your Existing Data

Your Database is Already Exported!

A complete snapshot of all 54 tables and all existing data (treks, batches, bookings, coupons, dropdowns, users, settings) has been exported to `gowild_database_backup.sql` (223 KB).

Step A: Copy the backup file from your laptop to the AWS server:

```
# Run this command on your local laptop terminal:

scp -i /path/to/gowild-key.pem \
    /var/www/html/goWILDKarunadu/gowild_database_backup.sql \
    ubuntu@YOUR_EC2_PUBLIC_IP:/home/ubuntu/
```

Step B: Create database and user on AWS server:

```
# On your AWS server terminal:

sudo mysql

CREATE DATABASE IF NOT EXISTS goWILDKarunadu CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci;

CREATE USER IF NOT EXISTS 'gowild_app'@'localhost' IDENTIFIED BY 'StrongPass@2026!';

GRANT ALL PRIVILEGES ON goWILDKarunadu.* TO 'gowild_app'@'localhost';

FLUSH PRIVILEGES;

EXIT;
```

Step C: Import all tables and data into AWS MySQL:

```
# Import the complete database with one command:

mysql -u gowild_app -p'StrongPass@2026!' goWILDKarunadu < /home/ubuntu/gowild_database_backup.sql

# Verify import (should show 54 tables):

mysql -u gowild_app -p'StrongPass@2026!' goWILDKarunadu -e "SHOW TABLES; SELECT COUNT(*) AS total_treks FROM treks;"
```

5. Step 4: S3 Bucket & CloudFront CDN (Always Free Tier)

5.1 Create S3 Bucket

1. Open **AWS S3 → Create bucket**.
2. **Bucket name:** `gowild-karunadu-images`
3. **Region:** `ap-south-1` (Mumbai).
4. **Block Public Access:** Keep **ENABLED (Checked)**.
5. Click **Create bucket**.

5.2 Request Free SSL Certificate in ACM (us-east-1)

1. Switch AWS region dropdown to **US East (N. Virginia) us-east-1**.
2. Open **Certificate Manager (ACM) → Request public certificate**.
3. **Domain name:** `images.gowildkarunadu.online` and `*.gowildkarunadu.online`.
4. Validation: **DNS validation** → Click **Request**.
5. Copy the CNAME record name and value shown, and add it into your **Hostinger DNS Records**. (ACM will validate and issue the free certificate within 5 minutes).

5.3 Create CloudFront Distribution (1 TB/month Free)

1. Open **CloudFront Console → Create distribution**.
2. **Origin domain:** Select `gowild-karunadu-images.s3.ap-south-1.amazonaws.com`.
3. **Origin access:** Select **Origin access control settings (OAC)** → click **Create new OAC**.
4. **Viewer protocol policy:** Select **Redirect HTTP to HTTPS**.
5. **Alternate domain name (CNAME):** `images.gowildkarunadu.online`.
6. **Custom SSL certificate:** Select the free ACM certificate from step 5.2.
7. Click **Create distribution**.
8. Copy the CloudFront domain name (e.g. `d123456.cloudfront.net`) and paste it into the `images` CNAME record in Hostinger DNS!
9. Copy the policy from the yellow banner, open your S3 bucket → **Permissions** → **Bucket Policy** → **Edit** →

Paste and **Save**.

6. Step 5: Deploy Code & Start Backends with PM2

```
# Clone repository

sudo mkdir -p /var/www/gowild

sudo chown -R ubuntu:ubuntu /var/www/gowild

cd /var/www/gowild

git clone <YOUR_GITHUB_REPO_URL> .
```

6.1 User API Environment (/var/www/gowild/goWILDKarunadu-user-api/.env)

```
cd /var/www/gowild/goWILDKarunadu-user-api

npm install --omit=dev

nano .env

PORT=4002

DB_HOST=127.0.0.1

DB_PORT=3306

DB_USER=gowild_app

DB_PASSWORD=StrongPass@2026!

DB_NAME=goWILDKarunadu


CLIENT_URL=https://gowildkarunadu.online

CORS_ORIGINS=https://gowildkarunadu.online,https://www.gowildkarunadu.online,https://
admin.gowildkarunadu.online

JWT_SECRET=your_jwt_secret_min_32_chars

CRYPTO_SECRET=JagguBoss_Secret_2025!


# Free Tier S3 + CloudFront

STORAGE_MODE=s3

AWS_REGION=ap-south-1
```

```
S3_BUCKET=gowild-karunadu-images

ADMIN_AWS_ACCESS_KEY=YOUR_AWS_IAM_KEY

ADMIN_AWS_SECRET_KEY=YOUR_AWS_IAM_SECRET

CUSTOM_CDN_URL=https://images.gowildkarunadu.online

CLOUDFRONT_URL=https://images.gowildkarunadu.online
```

6.2 Admin API Environment (/var/www/gowild/goWILDKarunadu-admin-api/.env)

```
cd /var/www/gowild/goWILDKarunadu-admin-api

npm install --omit=dev

nano .env

PORT=4001

DB_HOST=127.0.0.1

DB_PORT=3306

DB_USER=gowild_app

DB_PASSWORD=StrongPass@2026!

DB_NAME=goWILDKarunadu


CORS_ORIGINS=https://gowildkarunadu.online,https://www.gowildkarunadu.online,https://
admin.gowildkarunadu.online

ADMIN_SOCKET_URL=https://admin-api.gowildkarunadu.online

USER_API_URL=http://127.0.0.1:4002

JWT_SECRET=your_jwt_secret_min_32_chars

RESPONSE_ENCRYPTION_KEY=JagguBoss_Secret_2025!

ENCRYPTION_SALT=start-here-salt


# Free Tier S3 + CloudFront

STORAGE_MODE=s3

AWS_REGION=ap-south-1

S3_BUCKET=gowild-karunadu-images
```

```
ADMIN_AWS_ACCESS_KEY=YOUR_AWS_IAM_KEY

ADMIN_AWS_SECRET_KEY=YOUR_AWS_IAM_SECRET

CUSTOM_CDN_URL=https://images.gowildkarunadu.online

CLOUDFRONT_URL=https://images.gowildkarunadu.online
```

6.3 Start Backends with PM2 (24/7 Cloud Execution)

```
# Start user API

cd /var/www/gowild/goWILDKarunadu-user-api

pm2 start app.js --name "gowild-user-api"


# Start admin API

cd /var/www/gowild/goWILDKarunadu-admin-api

pm2 start app.js --name "gowild-admin-api"


# Make PM2 restart automatically if instance ever reboots

pm2 save

pm2 startup
```

7. Step 6: Build Frontends & Configure Nginx

```
# Build User Angular UI (using 4GB swap space)

cd /var/www/gowild/goWILDKarunadu-user-ui

npm install && npx ng build --configuration production


# Build Admin Angular UI

cd /var/www/gowild/goWILDKarunadu-admin-ui

npm install && npx ng build --configuration production
```

7.1 Nginx Reverse Proxy Configuration (/etc/nginx/sites-available/gowild.conf)

```
sudo nano /etc/nginx/sites-available/gowild.conf
```

1. USER FRONTEND (gowildkarunadu.online)

```
server {  
  
    server_name gowildkarunadu.online www.gowildkarunadu.online;  
  
    root /var/www/gowild/goWILDKarunadu-user-ui/www;  
  
    index index.html;  
  
    location / {  
  
        try_files $uri $uri/ /index.html;  
  
    }  
  
}
```

2. ADMIN FRONTEND (admin.gowildkarunadu.online)

```
server {  
  
    server_name admin.gowildkarunadu.online;  
  
    root /var/www/gowild/goWILDKarunadu-admin-ui/www;  
  
    index index.html;  
  
    location / {  
  
        try_files $uri $uri/ /index.html;  
  
    }  
  
}
```

3. USER API (api.gowildkarunadu.online -> Node port 4002)

```
server {  
  
    server_name api.gowildkarunadu.online;  
  
    location / {  
  
        proxy_pass http://127.0.0.1:4002;  
  
        proxy_http_version 1.1;  
  
        proxy_set_header Upgrade $http_upgrade;  
  
        proxy_set_header Connection 'upgrade';  
  
    }  
  
}
```

```

    proxy_set_header Host $host;

    proxy_set_header X-Real-IP $remote_addr;

    proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;

    proxy_set_header X-Forwarded-Proto $scheme;

    client_max_body_size 50M;

}

}

```

4. ADMIN API (admin-api.gowildkarunadu.online -> Node port 4001)

```

server {

    server_name admin-api.gowildkarunadu.online;

    location / {

        proxy_pass http://127.0.0.1:4001;

        proxy_http_version 1.1;

        proxy_set_header Upgrade $http_upgrade;

        proxy_set_header Connection 'upgrade';

        proxy_set_header Host $host;

        proxy_set_header X-Real-IP $remote_addr;

        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;

        proxy_set_header X-Forwarded-Proto $scheme;

        client_max_body_size 50M;

    }

}

```

Enable and restart Nginx:

```

sudo ln -sf /etc/nginx/sites-available/gowild.conf /etc/nginx/sites-enabled/

sudo rm -f /etc/nginx/sites-enabled/default

sudo nginx -t

```

```
sudo systemctl reload nginx
```

7.2 Activate Free SSL with Certbot (Let's Encrypt)

```
sudo certbot --nginx \  
-d gowildkarunadu.online \  
-d www.gowildkarunadu.online \  
-d admin.gowildkarunadu.online \  
-d api.gowildkarunadu.online \  
-d admin-api.gowildkarunadu.online
```

8. Step 7: Push Local Images to S3 & Update Database

```
cd /var/www/gowild/goWILDKarunadu-admin-api  
  
node scripts/sync-local-uploads-to-s3.js --update-db
```

9. Updating Code in Future (Zero Downtime)

1. On your laptop: `git push origin main`
2. On the AWS EC2 server:

```
cd /var/www/gowild  
  
git pull  
  
# If frontends changed:  
  
cd goWILDKarunadu-user-ui && npx ng build --configuration production  
  
cd ../goWILDKarunadu-admin-ui && npx ng build --configuration production  
  
# If backends changed:  
  
pm2 restart all
```